PERSONAL SYSTEM REQUEST-FLOW DIAGRAM

Group name-Apex Unified

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https://youtu.be/soQgutppeiY?si=sLuXBXmr_VQwD1us

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Typed Explanation of the HavenRae Project Management Platform Request Flow

The request flow diagram for the HavenRae Project Management Platform illustrates how a request moves from the user interface to the backend server, database, and back to the user. The example request used is creating a new project, which is one of the core functions of the platform. This explanation describes each stage of the process and answers the required system design questions based on the team's proposed implementation.

1. What device is the user on?

The primary users of the HavenRae Project Management Platform are expected to access the system using desktop or laptop computers. These include the Operations Manager, Interior Designers, Procurement Staff, Project Assistants, Management personnel and optional clients. Responsive web design will also support tablets. Native mobile applications are not planned for this semester, although they may be considered in future versions.The primary users of the HavenRae Project Management Platform are expected to access the system using desktop or laptop computers. These include the Operations Manager, Interior Designers, Procurement Staff, Project Assistants, Management personnel and optional clients. Responsive web design will also support tablets. Native mobile applications are not planned for this semester, although they may be considered in future versions.

2. Is it a browser or mobile app?

The HavenRae system will be a browser-based responsive web application rather than a native mobile application. Users will access it through modern web browsers. The frontend is planned to be built with React, Vite and Tailwind CSS.The HavenRae system will be a browser-based responsive web application rather than a native mobile application. Users will access it through modern web browsers. The frontend is planned to be built with React, Vite and Tailwind CSS.

3. What URL, route, screen or endpoint is involved?

The request begins on the New Project page. After the Operations Manager completes the form, the browser sends a POST request to /api/projects. When successful, the frontend redirects the user to the newly created project dashboard.The request begins on the New Project page. After the Operations Manager completes the form, the browser sends a

POST request to /api/projects. When successful, the frontend redirects the user to the newly created project dashboard.

4. What protocol is used?

Communication uses HTTPS with TLS encryption after DNS resolves the application domain. This protects sensitive business information while it travels across the network.Communication uses HTTPS with TLS encryption after DNS resolves the application domain. This protects sensitive business information while it travels across the network.

5. What data is sent?

The request contains the project name, client name, budget, start and end dates, assigned designer and authentication token. Data is transmitted as JSON because the backend is planned to use Django REST Framework.The request contains the project name, client name, budget, start and end dates, assigned designer and authentication token. Data is transmitted as JSON because the backend is planned to use Django REST Framework.

6. What validates the request?

The backend validates the JWT token, confirms the user's role, checks required fields, validates data types and ensures business rules are satisfied before creating the project.The backend validates the JWT token, confirms the user's role, checks required fields, validates data types and ensures business rules are satisfied before creating the project.

7. What database table, collection or external service is touched?

The request inserts a new record into the Projects table in PostgreSQL. User information may be referenced when assigning designers, while uploaded documents are planned to be stored in Cloudinary. Django ORM manages communication with the database.The request inserts a new record into the Projects table in PostgreSQL. User information may be referenced when assigning designers, while uploaded documents are planned to be stored in Cloudinary. Django ORM manages communication with the database.

8. What response comes back?

On success, the backend returns a JSON response containing a success message and the generated project ID. The frontend then redirects the user to the project dashboard and displays a confirmation message.On success, the backend returns a JSON response containing a success message and the generated project ID. The frontend then redirects the user to the project dashboard and displays a confirmation message.

9. What happens if validation fails?

The server rejects invalid requests and returns descriptive error messages. Inline validation messages are displayed while preserving the user's entered information.The server rejects invalid requests and returns descriptive error messages. Inline validation messages are displayed while preserving the user's entered information.

10. What happens if the database is down?

The application returns an internal server error, no partial records are created and the user is informed to try again later.The application returns an internal server error, no partial records are created and the user is informed to try again later.

11. What happens if 500 users do this at once?

Although such traffic is unlikely initially, Django and PostgreSQL can handle concurrent requests. Future improvements could include load balancing, additional backend instances and database optimization.Although such traffic is unlikely initially, Django and PostgreSQL can handle concurrent requests. Future improvements could include load balancing, additional backend instances and database optimization.

12. What security or privacy concern exists?

Sensitive client names, budgets, supplier information and project documents are protected through HTTPS, JWT authentication, role-based access control, secure Cloudinary storage and server-side validation. Sensitive information should never be exposed or logged in plain text.Sensitive client names, budgets, supplier information and project documents are protected through HTTPS, JWT authentication, role-based access control, secure Cloudinary storage and server-side validation. Sensitive information should never be exposed or logged in plain text.

AI DISCLOSURE STATEMENT

I used AI assistance

I understand that I must be able to explain and defend every part of this submission

Date	Tool	Prompt Summary	Output Used	Output Rejected	Verification
26 June 2026	ChatGPT	Requested assistance to organize the HavenRae request flow explanation, improve grammar, simplify technical	Used sentence restructuring, improved explanations, technical wording, and formatting suggestions to make the document more	Rejected any assumptions, features, or technical examples that did not match the team's HavenRae system design	Compared every AIgenerated response with the team's project proposal, request flow diagram, and planned technologies (React, Django REST Framework, PostgreSQL,
Date	**Tool**	**Prompt Summary**	**Output Used**	**Output Rejected**	**Verification**
		concepts, and ensure all required system design questions were clearly answered.	professional and easier to understand.	or project requirements.	Cloudinary, JWT, HTTPS). All content was edited to accurately reflect the team's implementation before submission.

REQUEST FLOW CHART

Github link-https://github.com/Dept-of-Comp-Sci-University-of-Ghana/dcit208-client-project-2026-team-engineering-repository-seg26-25-apexunited

Github personal link-https://github.com/Dept-of-Comp-Sci-University-of-Ghana/dcit208-client-project-2026-team-engineering-repository-seg26-25-apexunited/commits?author=Daryl-ac

Full Name: Acquaye Daryl Acquaye

Student ID: 22411960

Project Title: HavenRae Project Management

Team Name: Apex Unified (AU)

Date: June 26, 2026

Feature: Create a New Project

Possible Failure Point: The database could be unreachable, in which case no partial record is saved.

Security concern: sensitive Business data must not be logged in.